Alexander van Meegen

* 23 April 1991, Geldern, Germany • vanmeegen@physik.rwth-aachen.de
avmlab.physik.rwth-aachen.de • 2HpXCQ0AAAAJ
0000-0003-2766-3982

Theme: Understanding information processing in artificial and biological neural networks with statistical physics.

Professional experience (academic)

RWTH Aachen University **Aachen**
Assistant Professor (W1), Department of Physics. *Since 02/2026*

École Polytechnique Fédérale de Lausanne **Lausanne**
Scientist, Brain Mind Institute. Mentor: Prof. W. Gerstner. *09/2024–01/2026*

Harvard University **Cambridge, MA**
Swartz Fellow, Center for Brain Science. Mentor: Prof. H. Sompolinsky. *09/2022–08/2024*

Education

Jülich Research Center **Jülich**
PhD, Computational Neuroscience. Supervisor: Prof. S. J. van Albada. *03/2018–08/2022*
"Simulation and Theory of Large-Scale Cortical Networks" (summa cum laude; awarded by the University of Cologne).

Humboldt-Universität zu Berlin **Berlin**
Master of Science, Physics. Supervisor: Prof. B. Lindner. *04/2015–09/2017*

Université de Toulouse **Toulouse**
Erasmus exchange, Physics. *09/2015–01/2016*

Ruprecht-Karls-Universität Heidelberg **Heidelberg**
Bachelor of Science, Physics. Supervisor: Prof. U. Schwarz. *09/2011–03/2015*

Grants, scholarships, and prizes

NRW Return Program
Ministry of Culture and Science of North Rhine-Westphalia *2025–2030*
PI for €1,250,000 startup grant; topic of the call: "Future Computing"; one of three awardees.

Postdoctoral fellowship
Swartz Foundation *2022–2024*
Financial support for postdoctoral research in theoretical neuroscience (~$80,000/year); only awardee at Harvard.

Excellence Award
Jülich Research Center *2023*
€5,000 awarded to internationally successful young scientists for an outstanding dissertation; one of three awardees.

Doctoral scholarship
Studienstiftung des deutschen Volkes *2020–2022*
Academic and financial support for doctoral studies in computational neuroscience (~€22,000/year).

Weltwärts scholarship
Federal Ministry for Economic Cooperation and Development *2010*
Full financial support for a one-year voluntary service in Kenya.

Publications

Peer-reviewed journals

- A. van Meegen, H. Sompolinsky (2025): *Coding schemes in neural networks learning classification tasks*, Nature Communications 16, 3354.
- J. Stubenrauch, C. Keup, A. Kurth, M. Helias, A. van Meegen (2025): *Fixed point geometry in chaotic neural networks*, Phys. Rev. Research 7, 023203.
- F. Martinelli*, A. van Meegen*, B. Şimşek, W. Gerstner, J. Brea (2025): *Flat channels to infinity in neural loss landscapes*, NeurIPS 2025.
- D. G. Clark, O. Marschall, A. van Meegen, A. Litwin-Kumar (2025): *Connectivity structure and dynamics of nonlinear recurrent neural networks*, Phys. Rev. X 15, 041019.
- J. Pronold*, A. van Meegen*, R. O. Shimoura*, H. Vollenbröker, M. Senden, C. C. Hilgetag, R. Bakker, S. J. van Albada (2024): *Multi-scale spiking network model of human cerebral cortex*, Cereb. Cortex, bhae409.
- M. Dick, A. van Meegen, M. Helias (2024): *Linking network- and neuron-level correlations by renormalized field theory*, Phys. Rev. Research 6, 033264.
- A. Morales-Gregorio*, A. van Meegen*, S. J. van Albada (2023): *Ubiquitous lognormal distribution of neuron densities in mammalian cerebral cortex*, Cereb. Cortex, bhad160.
- K. Segadlo*, B. Epping*, A. van Meegen*, D. Dahmen, M. Krämer, M. Helias (2022): *Unified field theoretical approach to deep and recurrent neuronal networks*, J. Stat. Mech. 10, 103401.
- M. Layer, J. Senk, S. Essink, A. van Meegen, H. Bos, M. Helias (2022): *NNMT: mean-field based analysis tools for neuronal network models*, Front. Neuroinform. 16, 835657.
- A. van Meegen, T. Kühn, M. Helias (2021): *Large-deviation approach to random recurrent neuronal networks: parameter inference and fluctuation-induced transitions*, Phys. Rev. Lett. 127, 158302.
- A. van Meegen, S. J. van Albada (2021): *Microscopic theory of intrinsic timescales in spiking neural networks*, Phys. Rev. Research 3, 043077.
- A. van Meegen, B. Lindner (2018): *Self-consistent correlations of randomly coupled rotators in the asynchronous state*, Phys. Rev. Lett. 121, 258302.

*: Shared first authorship.

Book chapters

- S. J. van Albada, J. Pronold, A. van Meegen, M. Diesmann (2019): *Usage and scaling of an open-source spiking multi-area model of monkey cortex*, BrainComp 2019, Lecture Notes in Computer Science Vol. 12339, Springer.

Preprints

- L. Pezon and A. van Meegen (2026): *Interpretable compositional computation with recurrent neural networks*, bioRxiv 2026.06.23.733979.
- K. J. Sandbrink, F. Martinelli, A. van Meegen, W. Gerstner, J. Brea (2026): *Teacher geometry shapes learnability in teacher-student networks*, arXiv 2609.09595.

Teaching

Lectures and seminars

Summer 2026: Seminar "Current Topics in Deep Learning Theory", *RWTH Aachen*.

Spring 2024: Teaching fellow "Computational Neuroscience", *Harvard*.

Spring 2023: Teaching fellow "Statistical Mechanics of Spin Glasses and Neural Networks", *Harvard*.

2020, 2021 & 2022: Lectures for "Introduction to Computational Neuroscience" series, *RWTH Aachen*.

Summer 2020: Teaching assistant "Theoretical Neuroscience", *RWTH Aachen*.

Winter 2016/17: Teaching assistant "Statistical Physics", *HU Berlin*.

Summer schools and conferences

07/2019: Introduction to the NEST simulator, *CNS 2019*, Barcelona.

04/2019: NEST simulator tutorial, *EITN Spring School*, Paris.

02/2019: NEST simulator tutorial, *3rd HBP Student Conference*, Ghent.

03/2018: NEST & TVB project "From local circuits to whole-brain models", *EITN Spring School*, Paris.

Supervision

Independent supervision

Max Wollgarten: PhD student (since 03/2026), Physics, *RWTH Aachen*.

Maximilian Schwabe: PhD student (since 07/2026), Physics, *RWTH Aachen*.

Timm Vordermark: MSc thesis (since 04/2026), Physics, *RWTH Aachen*.

Maximilian Rechl: MSc thesis (since 10/2026), Physics, *RWTH Aachen*.

David Jänsch: MSc thesis (since 10/2026), Physics, *RWTH Aachen*.

Dimitar Martinov: BSc thesis (since 08/2026), Physics, *RWTH Aachen*.

Erik Martinek: BSc thesis (since 08/2026), Physics, *RWTH Aachen*.

Co-supervision

Emilio Scontrino: MSc thesis co-supervised with J. Erdmann (since 10/2026), Data Science, *RWTH Aachen*.

Charles-Edouard Rouault: Semester project in W. Gerstner's lab "How does sparsity shape the solution space of neural networks?" (2024–2025), Neuro-X, *EPFL*.

Grégoire Briolay: Semester project in W. Gerstner's lab "How do energy constraints shape the solution space of neural networks?" (2024–2025), Communication Systems, *EPFL*.

Jakob Stubenrauch: MSc thesis co-supervised with M. Helias "Phase Space Topology of Random Recurrent Neural Networks" (2021–2022), Physics, *RWTH Aachen*.

Kai Segadlo: MSc thesis co-supervised with M. Krämer and M. Helias "Theory of Learning and Prediction by Deep and Recurrent Networks in Gaussian Process Approximation" (2020–2021), Physics, *RWTH Aachen*.

Hannah Vollenbröker: MSc thesis co-supervised with S. van Albada "Simulation Studies of Biological Stabilization Mechanisms in Human Cerebral Cortex" (2019–2021), Translational Neuroscience, *HHU Düsseldorf*.

Bastian Epping: BSc thesis co-supervised with M. Krämer and M. Helias "Neural Networks as Gaussian Processes" (2020), Physics, *RWTH Aachen*.

Michael Dick: MSc thesis co-supervised with M. Helias "Renormalized Fluctuation Expansion for Non-Equilibrium Disordered Networks" (2019–2020), Physics, *RWTH Aachen*.

Georg Chechelnizki: Lab rotation in B. Lindner's lab (2016–2017), Computational Neuroscience, *HU Berlin*.

Selected talks and posters

Invited talk: *Interpretable representations for classification tasks*, AXIOM Workshop, NeurIPS 2026, Paris.

Invited talk: *Flexible computation with shared latent dynamics in low-rank RNNs*, Bernoulli Workshop, Lausanne.

Poster: *Hallmarks of computation via shared latent dynamical motifs*, COSYNE 2026, Lisbon.

Invited talk: *Symmetries in neural loss landscapes and their consequences*, Bocconi Workshop, Milan.

Invited talk: *Coding schemes in neural networks learning classification tasks*, Neurophysics Seminar, Berlin.

Plenary talk: *State space structure of random recurrent neuronal networks*, COSYNE 2023, Montreal.

Invited talk: *Statistical physics approach to neuronal dynamics*, INTheory Online Seminar Series.

Poster: *Inferring random network parameters from continuous-time trajectories*, COSYNE 2020, Denver.
Plenary talk: *Self-consistent correlations of randomly coupled rotators*, CNS 2019, Barcelona.
Invited talk: *A bridge from large deviation theory to statistical field theory*, EITN Mean-Field Workshop, Paris.
Contributed talk: *Self-consistent correlations of randomly coupled rotators*, DPG Spring Meeting, Berlin.

Organization of conferences & workshops

Quo vadis, neural network theory? **Frankfurt**
Satellite workshop at Bernstein Conference; organized together with J. Zavatone-Veth. *09/2026*

4th HBP Student Conference **Pisa**
Scientific committee *01/2020*

3rd HBP Student Conference **Ghent**
Scientific committee *02/2019*

INM-6 Institute Retreat 2018 & 2019 **Heijen**
Organizing committee *05/2018 & 05/2019*

Peer reviewing

Ad hoc reviewer for Physical Review Letters, Nature Communications, Physical Review E, Biological Cybernetics, Frontiers in Neural Circuits, Communications Biology, PLOS ONE, and Entropy.

Memberships

Current: Bernstein Network Computational Neuroscience, German Physical Society (DPG).

Languages

German: Native speaker
English: Business proficient *Daily use, both written and spoken.*
French: Fluent *Semester abroad (France); lived in Lausanne for 1.5 years.*
Swahili: Intermediate *Worked in Kenya for one year.*
Dutch: Basic *Dutch family (stepfather).*

Professional experience (industry)

VDI/VDE Innovation + Technik GmbH **Berlin**
Working student *04/2016–08/2017*
Developed and implemented a text-mining framework for analyzing research and development databases as well as news feeds. Results were used for policy advice to the German Federal Ministry of Education and Research.

EWC Weather Consult GmbH **Karlsruhe**
Working student *10/2014–05/2015*
Developed and evaluated a statistical method to improve energy forecasts in solar farms.

EWC Weather Consult GmbH **Karlsruhe**
Internship *08/2013–10/2013*
Developed and evaluated a wake model to improve energy forecasts in wind farms.

Voluntary services

Jülich Research Center **Jülich**
PhD representative INM-6 *2019–2021*
Represented PhD students' interests at institute meetings, kept in touch with research center committees, and mediated in case of conflicts.

Heidelberg Institute for International Conflict Research e.V. **Heidelberg**
Voluntary staff *2013*
Researched the state of violent conflicts in Eritrea for the annual "Conflict Barometer".

ICJA Freiwilligenaustausch weltweit e.V. **Berlin**
Instructor, mentor, and chair of the regional group Baden-Württemberg *2012–2015*
Mentored international volunteers in Germany, organized monthly social meetings, and mediated in case of conflicts. Co-organized and co-conducted a final seminar for international volunteers in Germany.

CoWaRT and Kiptere.CH **Kiptere**
Voluntary service in Kenya *2010–2011*
- CoWaRT: HIV education in the Kiptere area.
- Kiptere.CH: Improvement of the socio-economic situation of young adults and farmers in the Kiptere area.